Graph Theory

Directions: Answer all the following questions. Turn your answers in as a single **PDF** file created with whatever word-processing or other document creation application you choose.

1. Give the adjacency matrix and adjacency list for:
    a. Figure 20-22
    b. Figure 20-34
2. Use both the depth-first and breadth-first strategies to traverse the graph in Figure 20-22 beginning with vertex 0. Do the same for Figure 20-36, beginning with vertex a. List the vertices in the order in which each traversal visits them.
3. For the graph 20-38:
    a. Draw all possible spanning trees.
    b. Draw the minimum spanning tree.
4. For Figure 20-22:
    a. Draw the DFS spanning tree whose root is vertex 0.
    b. Draw the BFS spanning tree whose root is vertex 0.

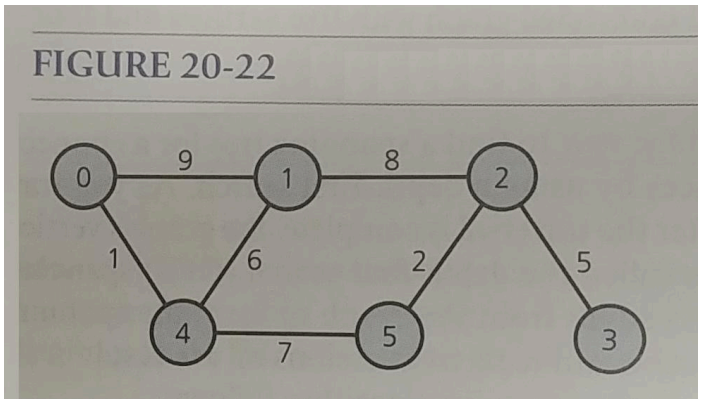

FIGURE 20-34 A graph for Exercise 1

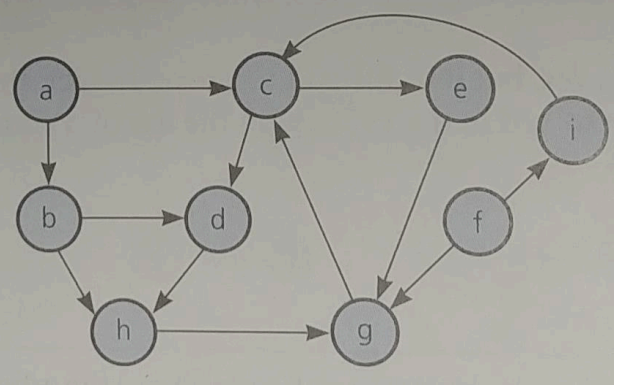

FIGURE 20-36 A graph for Exercises 4 and 10

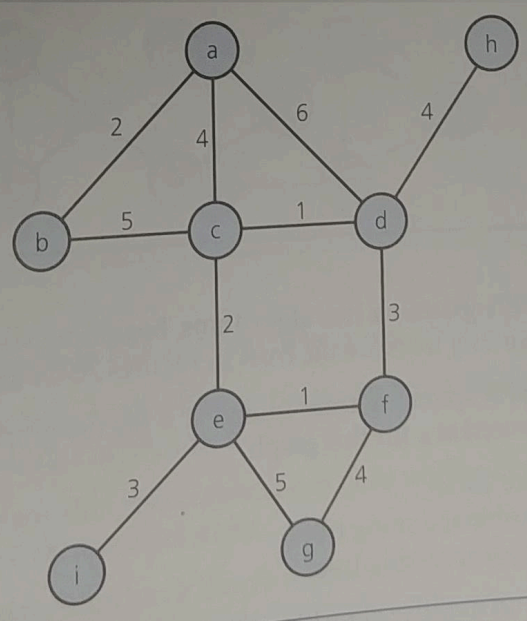
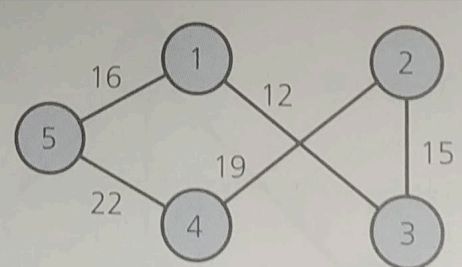
FIGURE 20-38 A graph for Exercise 11